The DQMH Framework

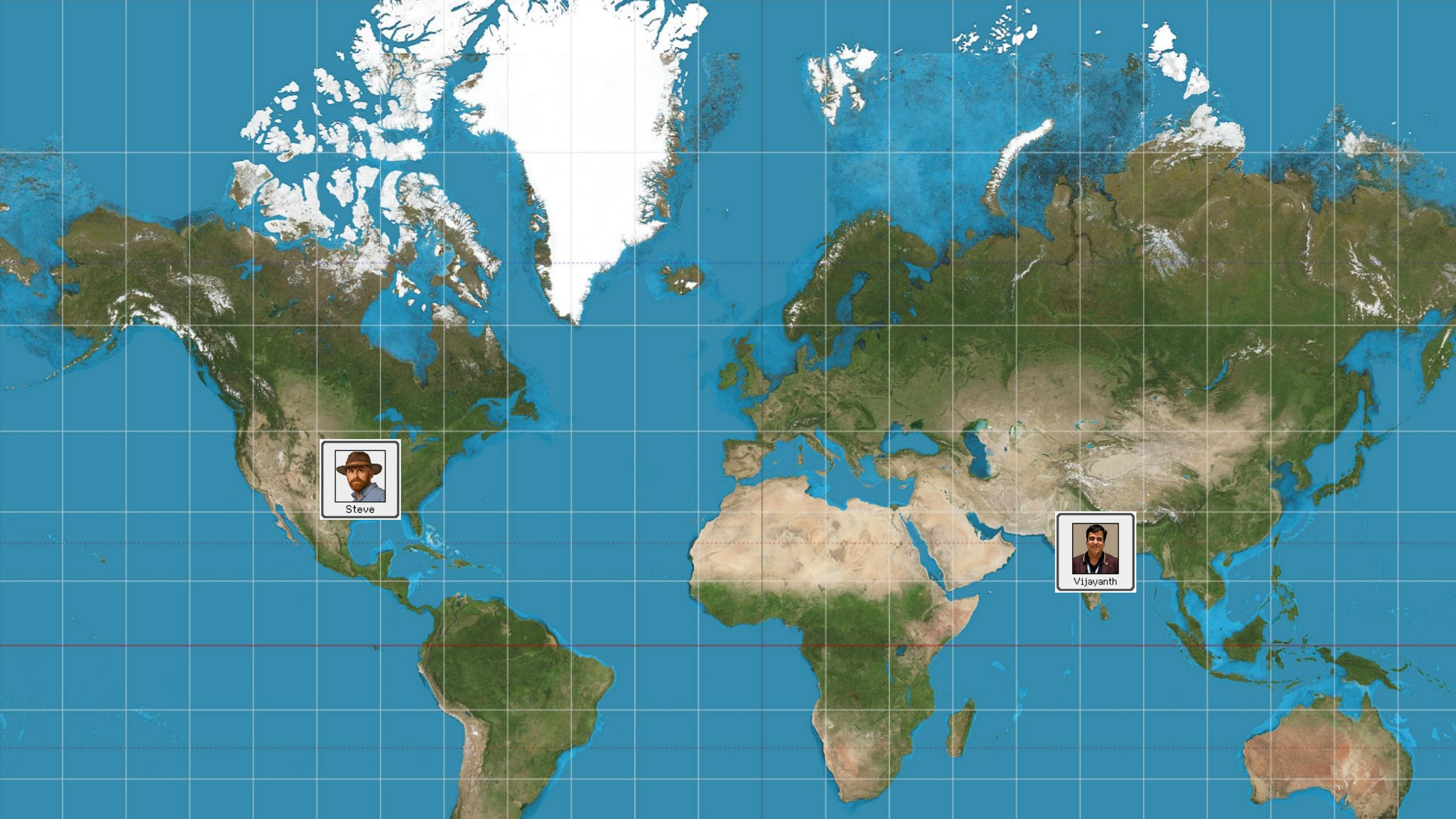
From Understanding to Execution

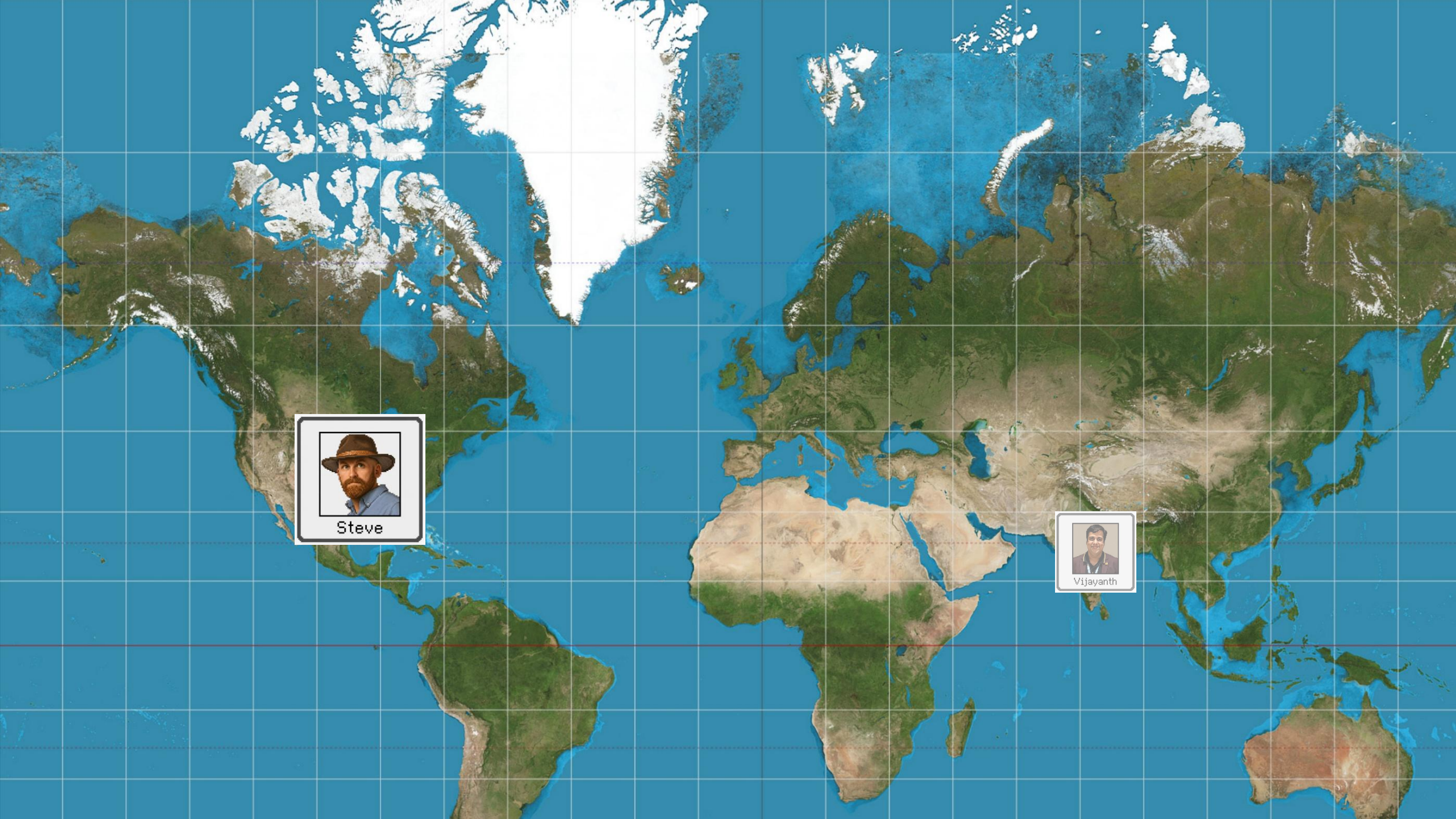


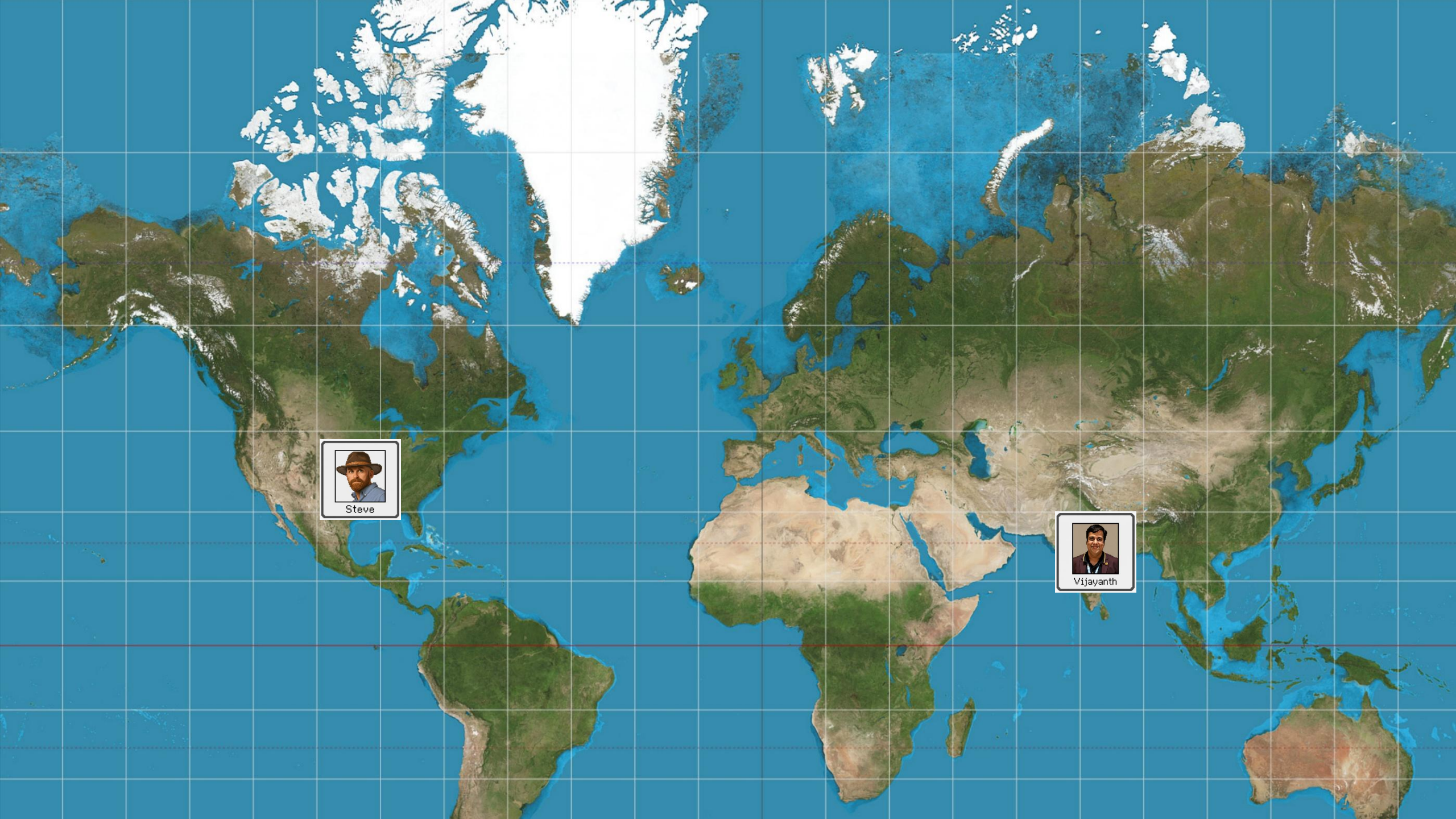


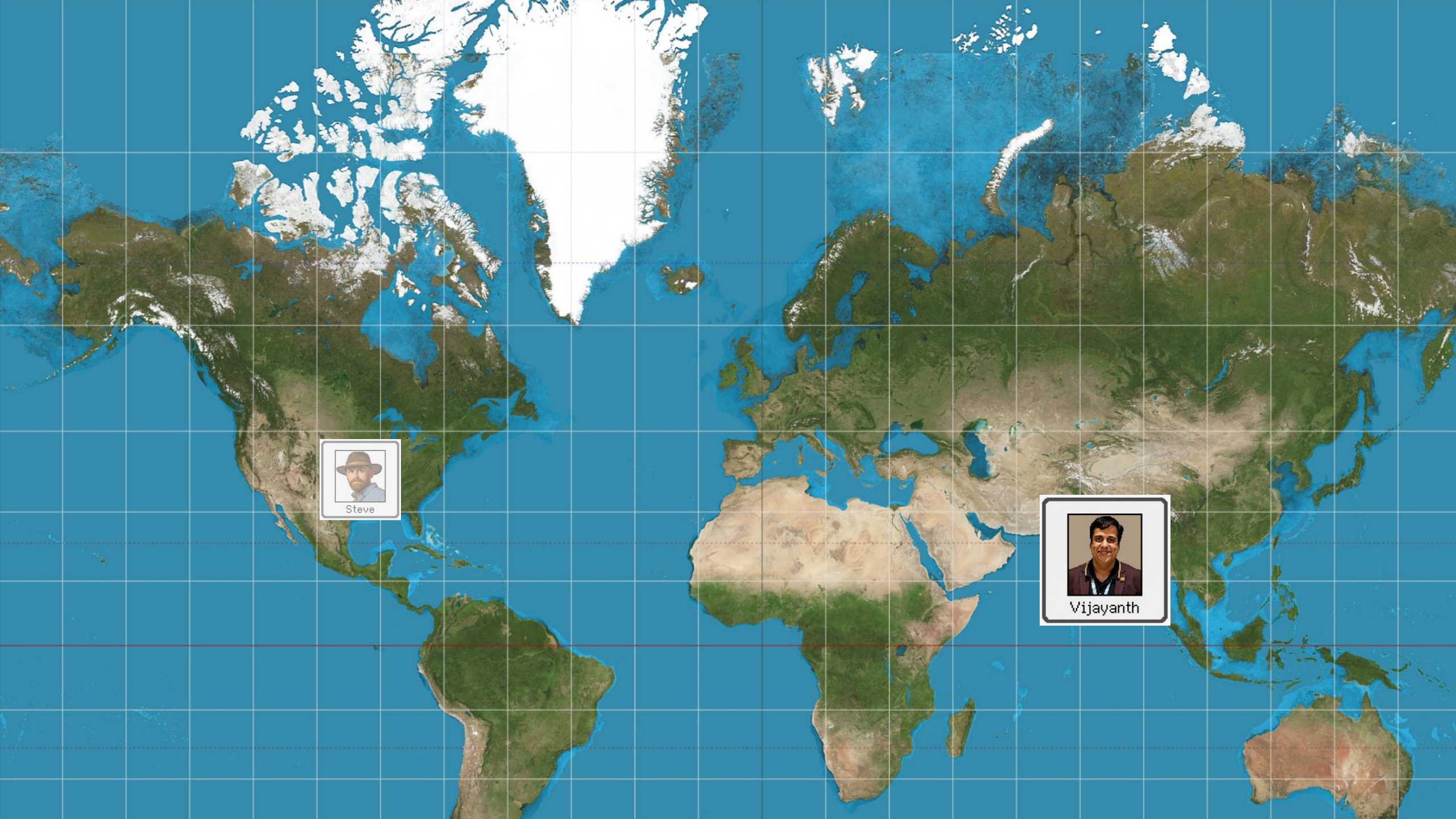
GLA Summit

August 31 - September 1, 2026


Steve
Vijayanth


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Vijayanth


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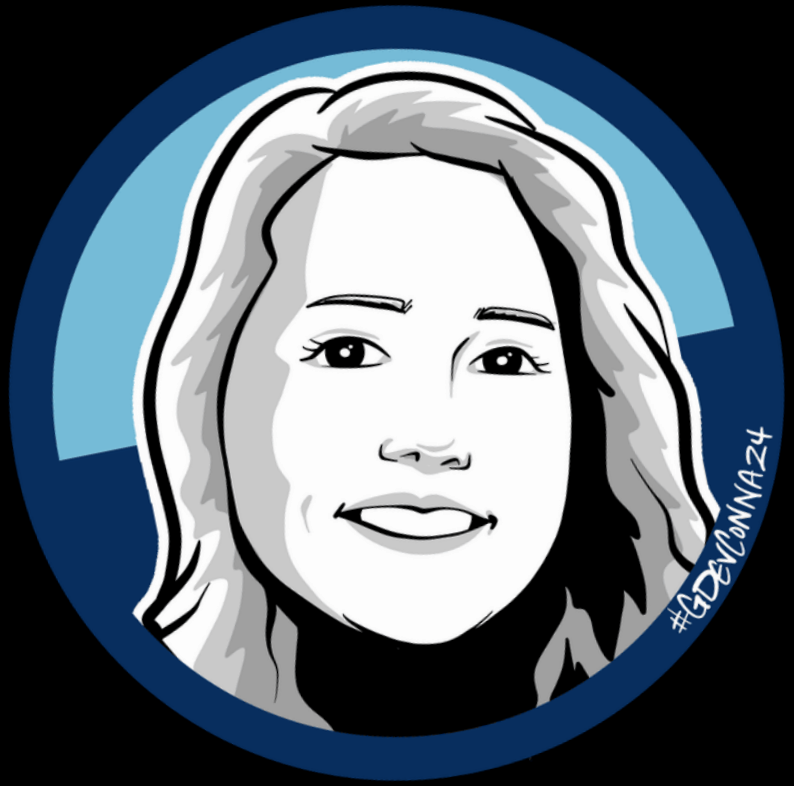

Steve
Vijayanth

#OurGiantsAreFemale

Fabiola De la Cueva



Nancy Henson



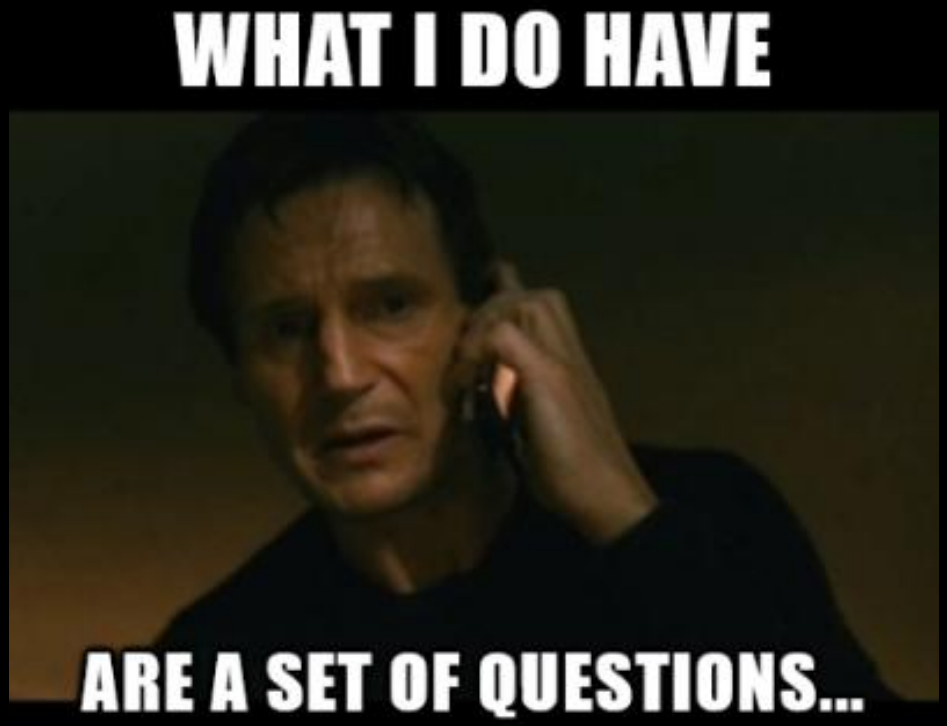
The DQMH Framework

- **D – Delacor**
- Framework because:
  - The boiler plate code is scripted for you
  - Guarantees a certain set of functions will happen unless you modify framework specific code
  - Allows users to focus directly on the business (functional) logic of the problem they seek to solve
  - Extensively documented
- **QMH – Queued Message Handler**
  - Business logic goes into the Message Handling Loop

Before We Get to DQMH…

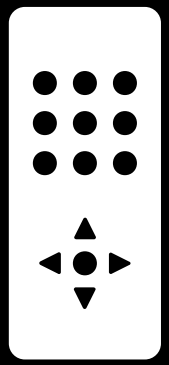


Over the Next Few Slides…



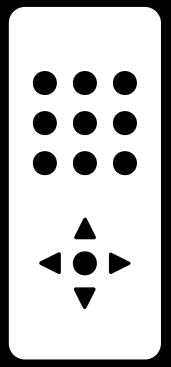
What Can You Do With the Remote?

- Perform Actions…
  - Turn on the TV
  - Change the channel
  - Decrease the volume
  - …



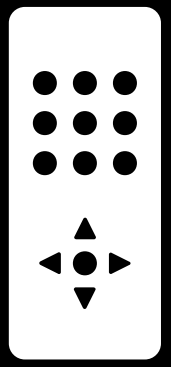
Does the Remote do all this?

- No!
- The TV is doing everything



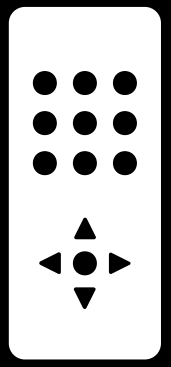
What are you ACTUALLY Doing With the Remote?

- Sending requests to the TV to perform actions
- Think of the remote as a collection of request senders
  - Each button is sending a different request to the TV



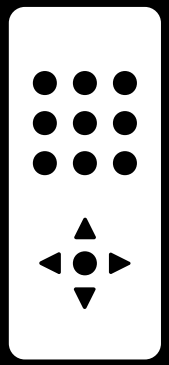
Does the Remote Model Matter?

- Yes
  - Try operating the TV with an AC remote



Does it matter who holds the remote?

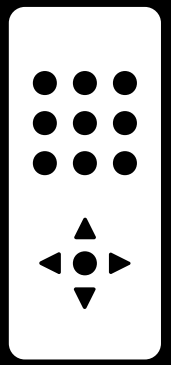
- No
  - The TV will respond the same way regardless



If two people held a remote each of the same make and model…

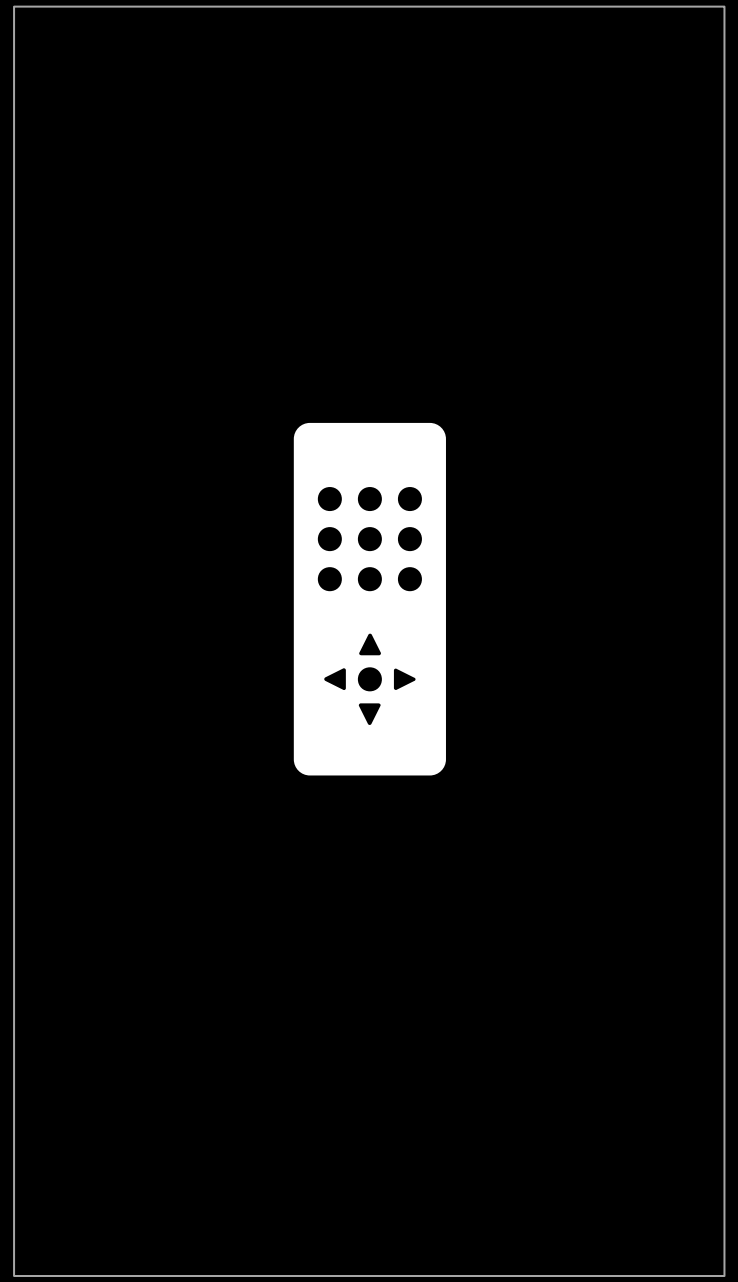
…And sent the same request to the TV, will the TV respond differently?

- Not if the TV is in the same state
- Requesting the TV to increase the volume from 17 will always increase it to 18, for e.g.



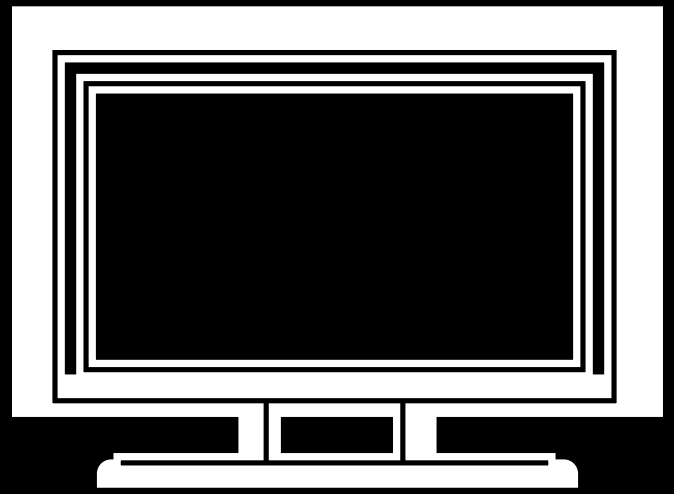
As a User of the TV and Remote…

- Do you need to be aware of the internal mechanics of how the TV performs its functions?
  - No!
  - The internal functionality is hidden from the user
- All you can do is “request” the TV to do something
- The TV responds only to the requests it is designed to accept
  - Sent via the remote
- Anyone in the public can send the request as long as they have the remote



What does the TV do?

- Broadcasts video and audio in sync
  - The mechanism of broadcasting, or the actual circuitry involved, is inside the TV
  - It is private to the TV
- Does the TV perform differently based on how many people are watching?
  - No!
  - Whoever is in line of sight of the TV can watch the TV
- Whoever is watching/hearing the TV can then request the TV to do something like increasing the volume



Any DQMH Module is like a TV…

- Other parts of the code can ask the DQMH module to perform actions by sending Requests
  - Requests are public
  - Can be sent by any part of the code
  - The module alone responds to the request

- The DQMH module can broadcast information
  - The actual broadcast generation is private
  - Other parts of the code can register for the broadcast user event and perform actions
  - The DQMH module itself does not care who is listening to the broadcasts

When designing a DQMH Module…

DO

- Think of the module functionality to implement
- Think of all the requests a DQMH module needs to respond to and how it will respond
- Think of what information will be broadcast by the DQMH module and when

DON’T

- Think of where the request is coming from
- What will be done with the broadcasted information

DO

- Keep the Tester up-to-date
- First work on the tester, then the code
- As you will see, the DQMH framework helps you do it this way

DEMO

- Creating DQMH Modules
- Testing what was created so far
- DQMH Events creation
- Integration of DQMH Modules into code

Resources

- Introduction to DQMH
- Introduction to DQMH - NI Community
- Everything But Occurrences, Darren Nattinger, Fabiola De la Cueva - GDevCon N.A. 2023
- LabVIEW Graphical Programming 5th edition
- DQMH Consortium Docs
- DQMH Project Templates and Examples

Confused about DQMH still? Ask Away!



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Discord: ughh_95001

Repo for code and reference material
https://github.com/Vijayanth-Reddy-K/GLA-Summit-2026-DQMH-Intro

Thank You!

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